API Documentation – QR/Intent

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Introduction:

UPI is a set of APIs created by NPCI to facilitate online immediate payments. UPI is expected to further propel easy instant payments via mobile. The payments can be both sender (payer) and receiver (payee) initiated and can be carried out using virtual payment addresses, Aadhaar integration, mobile number etc. The payer's smartphone could be used for secure credential capture.

Merchant on-boarding:

Merchant needs to provide the following information for onboarding of UAT and production environment:

Technical list:
- IP address (For dynamic IPs please provide range of IP addresses)
- Merchant call-back URL to post final transaction status from ICICI's end
- Merchant certificate with 4096 bits public key (.pem or .cer format) for encryption ➢ Merchant SSL certificate for sending call back response on call back url

Once the merchant provides all the above mentioned technical list, Bank will do the necessary configuration at their end and provide Merchant ID (MID) which shall be configured against the Virtual Payment Address (VPA). Once these details are received at merchant's end, they can start the API testing.

Bank will also provide ICICI bank's public key certificate for encryption to be done at merchant's end. Merchant will need to make encrypted request call using ICICI Bank's public key certificate to selected APIs from their Application Server and ICICI Bank will post encrypted response packet using merchant's public key certificate. Merchant is required to decrypt the response packet received at their end with the corresponding private key.

Encryption & Decryption Process: **ASYMMETRIC**

Asymmetric Encryption & Decryption Process Payload content-type will be in text/plain. Base64 encoded encrypted Cipher will be passed as a payload.

0G5mU1JJNBuwQaSLKb3wfRZks/cTV02yBNNuqjNHDWEC144W×C8iKqBpJAgq7reFKC4sHNUmNPRD ya1AvmQ7x1L
+3EAdEs9FEWNurZuWTvZpk4y7JrGhg0rz9KptBf+JfJUkSMo7NR3Saxel6EYtckkDr3AG
W7WJZmhcEoAMMXRws/hLVmaNHC/nOjCNqqBd4100AzdJh/HADRVI+YAJKT8dE4x9NTI+UX1zA00 Whza+
TsWEHfxzQla7zai7WSa/wiJD3uD7mk5vT1WY/fKJBquCuzM7135vigDhmb7dLVLuX8VMiNQrtErW NIOu Vaag 1jg +
uZUtyDSxjPFi5yEpKVVc7+ T5031DnCvkCFDygqasDsPL24qOjYk4XavTZvwGuPAdYNNkVn
L2VEIEhg42S2ye+fa/8fZiMt/3fwYeN9dgn9i5R6VOFbXSuZJYPSci9k0oqz73h1nzFtps60rUEDoGIkGvm9w
aJU3W78VH5mldGfGwJjiKlu/Hmi/huzEX9v4w3mW7RDGgmOuKImkqki +XWgyBOJvVmsLdO+cBaym/ seZP3+
zdfh09AWSI2DLD4Vf0jDjzoDSF2mzUFgHK9mbtbXgvsnReoGqx/KsivzmZNLmDmtg8eR4Z9Ln
Lni4r|401kDv5y/mxMtL3MBUUUajkw60S6NnhEG895yo

For encryption of request at ICICI:

Encrypted_Payload = Base64Encode(RSA/ECB/PKCS1Encryption(payload,ICICIPubKey.cer))

For decryption of response at Client:

plainResponse_ Base64Decode(RSA/ECB/PKCS1Decryption(encPayload, ClientPrivateKey.p12))

For encryption of response at ICICI:

Encrypted_Payload = Base64Encode(RSA/ECB/PKCS1Encryption(payload,ClientPubKey.cer))

General Flow:

1. For QR code or intent call transactions, merchant needs to send ‘refid’ in “tr” field of the QR/intent string.
2. There are two ways to generate ‘refid’.
   - First, is to call QR API. Merchant will send QR API request to QR API.On receiving request in correct format, ICICI bank will send QR API response which will have unique RefId starting with EZY or EZP prefix.
   - Second way, merchant will generate its own ‘refid’ with merchant specific three Prefix letter.
3. Using ‘refId’ merchant can create QR code or can initiate intent call. Customer will scan QR code or in case of intent call, customer will choose his PSP app to complete transaction.
4. Once Customer accepts or rejects the request from his mobile app, ICICI bank will send call back response to merchant stating Success or Reject on their callback URL.
5. When customer accepts the request from his mobile, transaction will be completed and amount will be credited to merchant’s account.
6. For ‘refid’ generated through QR API ICICI bank will send respective ‘merchantTranId’ of QR API request in call back response.
7. For merchant generated ‘refid’, ICICI bank will send merchant generated ‘refid’ in call back response except prefix letter.

API Details:

The specific name of each APIs are mentioned in the below sections. The customer parameters to be passed are specific to each API.

Below is the format for sending details.

First the parameters and their values will be entered in JSON Object. Then the whole JSON object will be encrypted and then encoded. Finally, the whole request will be passed through URL.

It will be a POST request

[GatewayURL(Base64Encode(RSA_Encrypt(JSON_Object{Field_Elements(field1,field2,…)})))]

The JSON Request Object is mentioned below where complete payload is encrypted using the public key provided by ICICI Bank: {

Base64Encode(RSA_Encrypt({ "merchantId" : "111111", "subMerchantId" : "12234", "terminalId" : "2342342", "merchantTranId" : "612413726581",…,… })) }

Encryption needs to be done using RSA 4,096 bits public key provided by ICICI Bank.

While sending the request please add the Headers in CODE which are Highlighted:

accept:*/*, accept-encoding:*, accept-language:en-US,en;q=0.8,hi;q=0.6, cache-control:no-cache, connection:keep-alive, content-length:684, content-type:text/plain;charset=UTF-8, host:apigwuat.icicibank.com:8443, origin:chrome-extension://fhbjgbiflinjbdggehcddcbncdddomop, postman-token:bfd89d8e-fd90-b9b7-b9da-17469eb99976, user-agent:Mozilla/5.0 (Windows NT 6.3; WOW64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/54.0.2840.71 Safari/537.36.

Below steps are to be followed:

1) Use Public certificate of ICICI Bank to encrypt payload using Algorithm RSA/ECB/PKCS1Padding Called Encrypted Payload
2) Base64 encode the Encrypted payload to get EncodedEncryptedPayload
3) The final string generated is the Request to API Gateway
4) Use content-type as Plaint/text in the body and call the API

Sample Payload:

X30i3+Y5kWiuQQ6/d+pW6oJaMidDpaXLznH03XUm6xRlUeAhKTghFb2SeXHzyNCkoi2+Ci8ms2OU
ljUhsJTyLWo+N6INqMNpki3ieQWBAxo+8s/xc9t/SSp3eLUIPgcEnwHJ93tDnvzD8KjRtWqo3mBg
ja84TnQvISM918WcUvZQi/NLbGjxlemBm2bHJYSfJwVtTMbJubvlZmAhrW14YpfY6B8ZzUBujZhf
qldMLL+B+zyKd9tlTztVCeVINQvDPhsNnU9OBNN+sHIESZzzi+B7PgYn7n/Mzpo594npZbZ9sDwS
XdwMlK1KY3rJXfzoq+RZL+dcl1ftfZjlqCHFfHposHzB3C3Smjm9EnzZEB0DfnxnT5CHvWM8l90I
1CKew9ZjKrbHAQ6y1eKDKad9935TlSh/WTirdtDHcpJW+HC8NYzd5lwzuVyWr24JD+riS7DcnKv7
YDH4xxnHaWXx/g8tgsxWK4H2m+VdvivVKWzAaX4GeNZd76uxGGUvKwxgqiyLasFqtYzYjOIm8fRq
jGpDdQ9CkrdmvyOOdOV+qFbXaMxCLyBAlrarFGO4QWoO5oJvmWY6zOXa/A2Apx+IX7CG51VuiwQZ
ssVEAGVzHQYn1n69nf2Jj/LLJXbg9gFg9naHTwf2m9jBorUhoo007Cm87v5oytwzGJIX13VOIAY=

1. API Name: QR API

Description: QR API will be used to fetch 'refid' from ICICI system. This 'refid' will be used to generate QR/ intent string.

As an optional feature, Validation of the Debit Account can be done using Validate Payer Acc Flag, Payer Account and Payer IFSC parameter.

UAT Endpoint: https://apibankingonesandbox.icicibank.com/api/MerchantAPI/UPI/v0/QR3/{merchantId}

Live Endpoint: https://apibankingone.icicibank.com/api/MerchantAPI/UPI/v0/QR3/{merchantId}

Input Parameters:

Name	Type	Description	Mandatory (Y-Yes / OOptional)	Length
merchantId	Number	Merchant Identification Number	Y	10
terminalId	Number	Needs to send Merchant category code (MCC code). [Default MCC-5411]	Y	4

amount	Number	Amount to be debited (In Rupees) in Integer value with 2 decimal E.g. : ‘200.00’ or ‘300.12’	Y	20
merchantTranId	AlphaNumeric	This will be a Transaction ID generated by the API and should always be unique	Y	35
billNumber	AlphaNumeric	Bill Number / Order Number	Y	50
validatePayerAccFl ag	String	‘Y’ for validating debit a/c details or ‘N’ for nonvalidation	O	1
payerAccount	Number	When ‘validatePayerAcc’ Flag is ‘Y’ then it is mandatory. Payer Account number is required to be entered.	O	
payerIFSC	AlphaNumeric	When ‘validatePayerAcc’ Flag is ‘Y’ then it is mandatory. Payer IFSC code is required to be entered.	O	

Sample Packet:


```
{
   "amount": "5.00",
   "merchantId": "118449",
   "terminalId": "5411",
   "merchantTranId": "p0nillp0k9lqlp091p17",
   "billNumber": "sdf1po111b",
   "validatePayerAccFlag": "Y",
   "payerAccount": "0405012740",
   "payerIFSC": "IC00000",
}
```


Output Parameters:

Name	Type	Description
merchantId	Number	Merchant Identification Number
terminalId	Number	Merchant category code
success	Alphanumeric	Result of the API Call

response	String	Response code indicating status of QR API
message	String	message description indicating status of QR API
merchantTranId	Alphanumeric	This will be a Unique Transaction ID generated by the Merchant.
refid	Alphanumeric	Reference id

<u>Response Packet:</u>


```
{
"response":"0",
"merchantId":"118449",
"terminalId":"5411",
"success":"true",
"message":"Transaction Initiated",
"merchantTranId":" p0nillp0k9lqlp091p17",
"refId":"EZY286844327832
}
```


<u>QR code generation string:</u>

upi://pay?pa=<merchant VPA>&pn=<merchant name>&tr=<Refid>&am=<amount>&cu=INR&mc=<MCC code>

<u>Example:</u> upi://pay?pa=abc@icici&pn=Abc&tr=EZY123456789012&am=10&cu=INR&mc=5411

2. API Name: Callback

<u>Description</u>: Final transaction response posted by ICICI Bank to Merchant's callback URL.

<u>Parameters:</u>

Name	Type	Description	Length
merchantId	Number	Merchant Identification Number	10
subMerchantId	Number	Sub Merchant Identification Number of Merchant	10
terminalId	Number	Merchant category code (MCC code). [Default MCC-5411]	10
BankRRN	Number	Bank reference Number for this transaction	20
merchantTranId	AlphaNumeric	Transaction Id as sent in the request packet	35

PayerName	AlphaNumeric	Name of the Payer	50
PayerMobile	Number	Mobile Number of the Payer	10
PayerVA	AlphaNumeric	Virtual Payment Address of the Payer	255
PayerAmount	Numeric	Amount of the Transaction	20
TxnStatus	AlphaNumeric	Status of the Transaction	20
TxnInitDate	Date and Time	Date and time on which the transaction was initiated	20
TxnCompletionDate	Date and Time	Date and time on which the transaction was completed	20

Sample Callback: Callback response after decryption.


```
{
  "merchantId" : "106161",
  "subMerchantId" : "12234",
  "terminalId" : "5411",
  "BankRRN" : "615519221396",
  "merchantTranId" : "612411454593",
  "PayerName" : "hhjjj",
  "PayerMobile" : "8879770059",
  "PayerVA" : "testing1@imobile",
  "PayerAmount" : "12",
  "TxnStatus" : "SUCCESS",
  "TxnInitDate" : "20160715142352",
  "TxnCompletionDate" : "20160715142352"
}
```


3. API Name: Transaction Status

Description: This API will be used by Merchant to get the status of the transaction based on 'merchantTranID' input parameter. This API will fetch the updated status from NPCI.

UAT Endpoint:
https://apibankingonesandbox.icicibank.com/api/MerchantAPI/UPI/v0/TransactionStatus3/{merchantId}

Live Endpoint:
https://apibankingone.icicibank.com/api/MerchantAPI/UPI/v0/TransactionStatus3/{merchantId}

Input Parameters:

Name	Type	Description	Mandatory (Y/N)	Length
merchantId	Number	Merchant Identification Number	Y	10
subMerchantId	Number	Sub Merchant Identification Number of Merchant	Y	10
terminalId	Number	Needs to send Merchant category code (MCC code). [Default MCC-5411]	Y	4
merchantTranId	AlphaNumeric	This will be a Transaction ID generated at the time of original request	Y	35

Sample Packet


```
{
  "merchantId": "118449",
  "subMerchantId": "118449",
  "terminalId": "5411",
  "merchantTranId": "p0nillp0k9lqlp091p17"
}
```


Output Parameters:

Name	Type	Description
Response	Number	Response Code
X`	Number	Merchant Identification Number
subMerchantId	Number	Sub Merchant Identification Number of Merchant
terminalId	Number	MCC
Success	String	Result of the API Call
Message	String	Response Code Description
merchantTranId	AlphaNumeric	This will be a Unique Transaction ID generated by the Merchant.
OriginalBankRRN	Number	Reference Number generated by Bank
Amount	Number	Amount
Status	AlphaNumeric	Status of the transaction

Sample Response:


```
{
  "response" : "0",
  "merchantId" : "106161",
  "subMerchantId" : "12234",
  "terminalId" : "5411",
  "OriginalBankRRN" : "615519221396",
  "merchantTranId" : "612411454593",
  "amount" : "12",
  "success" : "true",
  "message" : "Transaction Successful",
  "status" : "SUCCESS"
}
```


Current response:
PENDING, SUCCESS, FAILURE

4. API Name: Callback Status

Description: This API will be used by Merchant to get the status of the transaction by passing correct transaction type. This API will fetch the status of the transaction based on RRN or merchant transaction ID or ref-id from ICICI Switch.

UAT Endpoint:
https://apibankingonesandbox.icicibank.com/api/MerchantAPI/UPI/v0/CallbackStatus2/{merchantId}

Live Endpoint:
https://apibankingone.icicibank.com/api/MerchantAPI/UPI/v0/CallbackStatus2/{merchantId}

Input Parameters:

Name	Type	Description	Mandatory (Y/C)	Length
merchantId	Number	Merchant Identification Number	Y	10
subMerchantId	Number	Sub Merchant Identification Number of Merchant	Y	10
terminalId	Number	Needs to send Merchant category code (MCC code). [Default MCC-5411]	Y	4

transactionType	Alphabet	Flag to identify type of original transaction as C, R, Q or P as per below mentioned *table	Y	1
merchantTranId	AlphaNumeric	This will be a Transaction ID generated at the time of original request.	C	35
transactionDate	Date	Date of the Transaction	C	20
BankRRN	Number	Bank Reference Number of the original transaction	C	15
refID	AlphaNumeric	Reference Number passed in QR/Intent Call	C	

*Transaction Type Flag	Scenario	MerchantTranId	BankRRN	RefId	Transaction Date
C	Collect Pay Transaction	Mandatory	Non-Mandatory	N/A	N/A
R	Refund Transactions	Mandatory	Non-Mandatory	N/A	N/A
Q	ICICI generated ref-Id (EZY or EZP)	Mandatory	Non-Mandatory	Non-Mandatory	N/A
P	Merchant generated refId or Push	Any 1 out of 3	Any 1 out of 3	Any 1 out of 3	Mandatory when search based on Ref Id

Sample Packet:


```
{
  "merchantId": "118449",
  "subMerchantId": "118449",
  "terminalId": "5411",
  "transactionType": "C",
  "merchantTranId": "p0nillp0k9lqlp091p17"
}
```


Output Parameters:

Name	Type	Description
Response	Number	Response Code
merchantId	Number	Merchant Identification Number
subMerchantId	Number	Sub Merchant Identification Number of Merchant

terminalId	Number	MCC
success	String	Result of the API Call
message	String	Response Code Description
merchantTranId	AlphaNumeric	This will be a Unique Transaction ID generated by the Merchant.
OriginalBankRRN	Number	Reference Number generated by Bank
PayerVA	AlphaNumeric	Virtual Payment Address of the Payer
Amount	Number	Amount
status	AlphaNumeric	Status of the transaction
TxnInitDate	Date and Time	Date and time on which the transaction was initiated
TxnCompletionDate	Date and Time	Date and time on which the transaction was completed

Sample Response:


```
{
  "response" : "0",
  "merchantId" : "106161",
  "subMerchantId" : "12234",
  "terminalId" : "5411",
  "OriginalBankRRN" : "615519221396",
  "merchantTranId" : "612411454593",
  "Amount" : "12",
  "payerVA" : " testing1@imobile ",
  "success" : "true",
  "message" : "Transaction Successful",
  "status" : "SUCCESS",
  "TxnInitDate" : "20160715142352",
  "TxnCompletionDate" : "20160715142352"
}
```


5. API Name: Refund API

Description: This API needs to be used by Merchants to initiate refunds of the transactions. Both offline and online refunds are supported in the same API.

UAT Endpoint:
https://apibankingonesandbox.icicibank.com/api/MerchantAPI/UPI/v0/Refund/{merchantId}

Live Endpoint:
https://apibankingone.icicibank.com/api/MerchantAPI/UPI/v0/Refund/{merchantId}

Input Parameters:

Name	Type	Description	Mandatory (Y/N)	Length
merchantId	Number	Merchant Identification Number	Y	10
subMerchantId	Number	Sub Merchant Identification Number of Merchant	Y	10
terminalId	Number	Needs to send Merchant category code (MCC code). [Default MCC-5411]	Y	4
originalBankRRN	String	Original Transaction Id	Y	15
merchantTranId	String	Refund Transaction Id	Y	35
originalmerchantTranId	AlphaNumeric	Merchant TranID of Refund transaction.	Y	35
refundAmount	Number	Amount to be debited.(In Rupees, Integer value with 2 decimal)E.g. : 200.00 / 300.12	Y	20
payeeVA	AlphaNumeric	Alias name with which the payee can be identified by his registered entity.	N	255
Note	AlphaNumeric	Remarks entered by the payer for his reference.	Y	50
onlineRefund	String	Refund request mode – Online or Offline refund – 'Y' for online refund and 'N' for Offline refund	Y	1

Sample Packet:


```
{
   "merchantId": “106092”,
   "subMerchantId": “12234”,
   "terminalId": “2342342”,
   "originalBankRRN": "622415338172",
   "merchantTranId": "88442047",
   "originalmerchantTranId": "202020202021",
   "payeeVA": "yatin@imobile",
   "refundAmount": "10.00",
   "note": "refund-request",
   "onlineRefund": "Y"
}
```


Output Parameters:

Name	Type	Description
Response	Number	Response Code
merchantId	Number	Merchant Identification Number
subMerchantId	Number	Sub Merchant Identification Number of Merchant
terminalId	Number	MCC
success	String	Result of the API Call
message	String	Response Code Description
merchantTranId	AlphaNumeric	This will be a Unique Transaction ID generated by the Merchant.
OriginalBankRRN	Number	Reference Number generated by Bank. For Online refund, new RRN will be generated. For Offline, Original RRN will be returned
status	AlphaNumeric	Status of the transaction

Sample Response


```
{
   "merchantId": “106092”,
   "subMerchantId": “12234”,
   "terminalId": “2342342”,
   "success": "true",
   "response": “0”,
   "status": "SUCCESS",
   "message": "Transaction Successful",
   "originalBankRRN": "622415338172",
   "merchantTranId": "88442055"
}
```


6. Error Codes

Code	Description	Reasons
500	Internal Server Error	Internal Server Error
401	Unauthorized	APIkey,IP whitelisting or SSL not present
403	Forbidden	Request not proper.
429	Too Many Requests	Too Many Requests
8002	INVALID_JSON.	INVALID_JSON.

8003	INVALID_FIELD FORMAT OR LENGTH	Field is not in the format mentioned
8004	MISSING_REQUIRED_FIELD	Mandatory field is missing
8006	INVALID_FIELD_LENGTH	Length of field exceeds defined length
8007	Invalid JSON, OPEN CURLY BRACE MISSING	Open Brace missing in JSON
8008	Invalid JSON,END CURLY BRACE MISSING	Closing Brace missing in JSON
8009	Internal Server Error	White space characters
8010	INTERNAL_SERVICE_FAILURE	The system had an internal exception
8011	BACKEND_HOST_NOT_FOUND	The Server referenced in the URL cannot be reached.
8012	BACKEND_CONNECTION_TIMEOUT	Cannot connect to service
8013	BACKEND_READ_TIMEOUT	Cannot read from service
8014	BACKEND_BAD_URL	The URL is incorrect.
8017	INVALID JSON	Improper JSON
8016	Decryption Fail	Request not properly Encrypted
5000	Invalid Request	if the request is failed with some other reasons
5001	Invalid Merchant ID	If the merchant Id is not valid
5002	Duplicate Merchant TranId	Transaction is already initiated with merchant transaction id
5003	Merchant Transaction Id is mandatory	If merchant transaction id null
5004	Invalid Data	invalid packet
5005	Collect By date should be greater than or equal to Current date	If given collect by date is less than current date
5006	Merchant TranId is not available	No transaction initiated with given transaction id based on merchant id
5007	Virtual address not present	If merchant entered invalid customer VPA
5008	PSP is not registered	If merchant entered wrong PSP handler

5009	Service unavailable. Please try later.	Default error response for unexpected internal failures.
5010	Technical Error	If any technical error.
5011	Duplicate merchant TranId	Transaction is already initiated with merchant transaction id
5012	Request has already been initiated for this transaction	If request is initiated already for this transaction.
5013	Invalid VPA	If VPA does not exits
5014	Insufficient amount	If Original amount is less than refund amount
5015	Invalid Original TranId	If original transaction Id is not available
5016	Payee VA should not be Merchant VA	Should not be Merchant Virtual Address
5017	Sorry you can't initiate refund request	Merchant can initiate online refund only if online refund flag is enabled
5018	Merchant VPA and Reference ID is not match	
5019	Invalid Terminal Id	
5020	No response from Beneficiary Bank. Please wait for recon before initiating the transaction again.	For Deemed approved transactions or timed out requests
5021	Transaction Timed out. Please check transaction status before initiating again.	OSB Timed out for collect request
5022	Terminal Id is mandatory	
5023	Multiple transactions against given parameter. Please provide bank RRN	
5024	Record not found against given parameters	
5025	Please enter valid refund amount	
5026	Invalid Consumer number	
5027	Invalid merchant prefix	
5028	Virtual Address Already Exists	
5029	No Response From Switch	

5030	Please try again	In case Check VPA return actCode 950 from Switch
5031	Validity start date should not be less than current date	If Validity start date is less than current date
5032	Validity end date should not be less than validity start date	If Validity end date is less than validity start date
5033	Mandate request not created	Without initiating the manage mandate
5034	No Approved Mandates are available	If manage mandate request are not in SUCCESS state
5035	Mandate expired	If mandate validity period is completed
5036	Mandate amounts mis-matched	If manage mandate is EXACT and different amount given in execute mandate
5037	Execution amount exceeded to Mandate approved amount	If manage mandate is MAX and execution amount crossed in execute mandate
5038	Invalid Validate Payer Account Flag	If validate payer account flag is other than Y and N
5039	Invalid Payer Account	If Payer Account is null, empty or invalid pattern
5040	Invalid Payer IFSC	If Payer IFSC is null, empty or invalid pattern
5041	Invalid Sequence Number	
5042	Duplicate Sequence Number	
5043	Invalid Unique Merchant ReferenceId	
5044	Invalid Merchant Name	
5045	Invalid Marketing Name	
5046	Invalid Bank Assigned MerchantId	
5047	OSB Timeout	
5048	New Unique Merchant ReferenceId	
5049	Failed at switch. Please try registering again.	
5050	Details of Bank Assigned MerchantId not found	

5051	Duplicate Unique Merchant ReferenceId	

Encryption & Decryption Process: **ASYMMETRIC**

Asymmetric Encryption & Decryption Process Payload content-type will be in text/plain. Base64 encoded encrypted Cipher will be passed as a payload.

0G5mU1JJNBuwQaSLKb3wfRZks/cTV02yBNNuqjNHDWEC144W×C8iKqBpJAgq7reFKC4sHNUmNPRD ya1AvmQ7x1L
+3EAdEs9FEWNurZuWTvZpk4y7JrGhg0rz9KptBf+JfJUkSMo7NR3Saxel6EYtckkDr3AG
W7WJZmhcEoAMMXRws/hLVmaNHC/nOjCNqqBd4100AzdJh/HADRVI+YAJKT8dE4x9NTI+UX1zA00 Whza+
TsWEHfxzQla7zai7WSa/wiJD3uD7mk5vT1WY/fKJBquCuzM7135vigDhmb7dLVLuX8VMiNQrtErW NIOu Vaag 1jg +
uZUtyDSxjPFi5yEpKVVc7+ T5031DnCvkCFDygqasDsPL24qOjYk4XavTZvwGuPAdYNNkVn
L2VEIEhg42S2ye+fa/8fZiMt/3fwYeN9dgn9i5R6VOFbXSuZJYPSci9k0oqz73h1nzFtps60rUEDoGIkGvm9w
aJU3W78VH5mldGfGwJjiKlu/Hmi/huzEX9v4w3mW7RDGgmOuKImkqki +XWgyBOJvVmsLdO+cBaym/ seZP3+
zdfh09AWSI2DLD4Vf0jDjzoDSF2mzUFgHK9mbtbXgvsnReoGqx/KsivzmZNLmDmtg8eR4Z9Ln
Lni4r|401kDv5y/mxMtL3MBUUUajkw60S6NnhEG895yo

For encryption of request at ICICI:

Encrypted_Payload = Base64Encode(RSA/ECB/PKCS1Encryption(payload,ICICIPubKey.cer))

For decryption of response at Client:

plainResponse_ Base64Decode(RSA/ECB/PKCS1Decryption(encPayload, ClientPrivateKey.p12))

For encryption of response at ICICI:

Encrypted_Payload = Base64Encode(RSA/ECB/PKCS1Encryption(payload,ClientPubKey.cer))

Encryption & Decryption Process: **HYBRID**

Security:

- API Key needs to be passed in every request in the header and merchant IP will also be required for IP whitelisting.
- API Key needs to be passed in the parameter name: apikey
- API request and response to Merchant is secured using advanced and agreed upon encryption algorithm agreed to maintain data confidentiality and integrity.
- API Gateway uses the standard authenticating and authorizing process for the incoming request from merchant and for maintaining the integrity and confidentiality we apply state of art Encryption/ Decryption algorithm.

Encryption & Decryption Process:

For Encryption of a payload at Client's end.

encryptedKey = Base64Encode(RSA/ECB/PKCS1Encryption(SesionKey,ICICIPubKey.cer))

Session key is nothing but randomly one time generated string of length 16 (OR 32).

encryptedData = Base64Encode(AES/CBC/PKCS5Padding(Response,SessionKey))

1. Generate 16-digit random number (session key). Say RANDOMNO.
2. Encrypt RANDOMNO using RSA/ECB/PKCS1Padding and encode using Base64. Say ENCR_KEY.
3. Perform AES/CBC/PKCS5Padding encryption on request payload using RANDOMNO as key and iv-initialization vector. Say ENCR_DATA.
4. Now client may choose to send IV in request from one of below two options.
   a. Send Base64 Encoded IV in "iv" tag. (Recommended Approach)
   b. Send IV as a part of ENCR_DATA itself.

```
bytes[] iv = IV;

bytes[] cipherText = symmetrically encrypted Bytes (step3) bytes[] concatB
= iv + cipherText;

ENCR_DATA = B64Encode(concatB);
```

5. Now in the complete request, Client needs to send encrypted request in below format.

```
{
    "requestId": "<request-id for tracking purpose>",
    "service": "AccountCreation",
    "encryptedKey":  "<ENCR_KEY>",
    "oaepHashingAlgorithm":  "NONE", "iv":
    "<IV>",
    "encryptedData":  "<ENCR_DATA>",
    "clientInfo": "",
    "optionalParam": ""
}
```

For Decryption of a response at Client's end.

IV= getFirst16Bytes(Base64Decode(encryptedData)

SessionKey =
Base64Decode(RSA/ECB/PKCS1Decryption(encryptedKey,ClientPrivateKey.p12,)) Session key is nothing but randomly generated string of length 16 (OR 32) .

Response = Base64Decode (AES/CBC/PKCS5Padding Decryption(encryptedData,SessionKey, IV))

1. Get the IV- Base64 decode the encryptedData and get first 16 bytes and rest is encryptedResponse.

```
bytes[] IV= getFirst16Bytes(Base64Decode(encryptedData)
```

2. Decrypt encryptedKey using algo (RSA/ECB/PKCS1Padding) and Client’s private key. sessionKey =

   B64Decode(RSA/ECB/PKCS1Decryption(encryptedKey,ClientPrivateKey.p12,))

3. Decrypt the response using algo AES/CBC/PKCS5Padding.

   Response = Base64Decode
   (AES/CBC/PKCS5Padding
   Decryption(encryptedData,SessionKey, IV))

4. You need to skip first 16 bytes of response, as it contains IV.